DEBRIEF · SIMULATED CASE

Malignant hypertherm

Fast-forwarded 5 min
19:45 → 24:45
No alarms

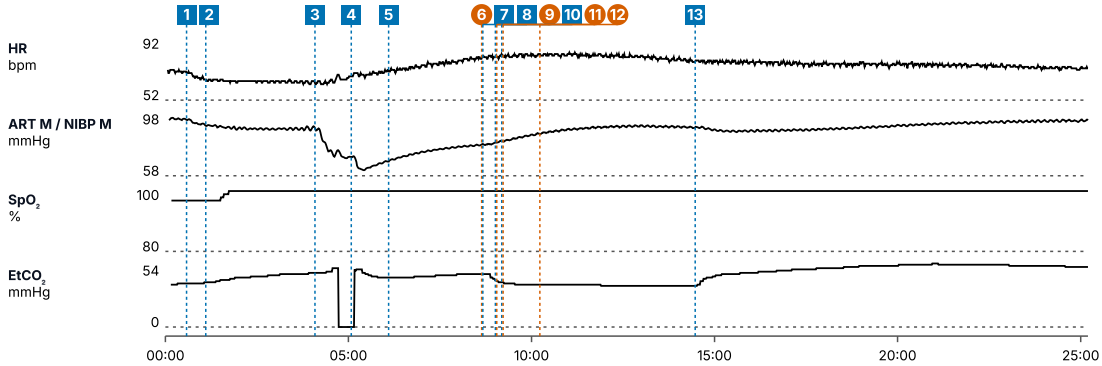
M 35 y 80 kg · 25:12 simulated (2 r , 12:59:06 AM · Run by: Live demo, 28 October

Key times

Recognition (instructor's tick)	Start → first team action	Call the MH crisis (stop sevoflurane, O_2 100 % at 10...	Give dantrolene 2.5 mg/kg within 300 s of calling the...
8 min 40 s 00:00 → 08:40	**35 s** 00:00 → 00:34 (Midazolam 2 mg and fentanyl 100 µg given, breathing room air)	**2 s** 08:38 → 08:40 · asked ≤ 2 min 00 s	**21 s** 08:40 → 09:00 · asked ≤ 5 min 00 s

The course

■ team action · ● patient event · □ scenario state · numbers match Key moments



Objectives

What the record shows: it reads the run, it does not grade the person

1. ✓ Observed — Name malignant hyperthermia from the monitor: $EtCO_2$ above 45 mmHg at the same ventilation, tachycardia (instructor-assessed)
   Observed by the instructor at 08:40.
2. ✓ Observed — Call the MH crisis (stop sevoflurane, O_2 100 % at 10 L/min, hyperventilate) within 120 s of the early signs ($EtCO_2$ above 45 mmHg at the same ventilation) 2 s of 2 min 00 s
   The state "Trigger off: sevoflurane stopped, O_2 100 % at 10 L/min, ventilation 20 × 600" 2 s after the state "Early signs: $EtCO_2$ rising at the same ventilation, tachycardia" (08:38 → 08:40; within 2 min 00 s asked).
3. ✓ Observed — Give dantrolene 2.5 mg/kg within 300 s of calling the crisis
   21 s of 5 min 00 s
   Dantrolene 21 s after the state "Trigger off: sevoflurane stopped, O_2 100 % at 10 L/min, ventilation 20 × 600" (08:40 → 09:00; within 5 min 00 s asked).
4. – Not assessable — Repeat dantrolene 2.5 mg/kg within 600 s of the first dose while $EtCO_2$ is still above 45 mmHg
   This objective has no rule: the instructor judges it.
5. – Not assessable — Treat the potassium (calcium chloride, insulin with dextrose) within 300 s of K^+ rising above 5.5 mmol/L
   K^+ above 5.5 mmol/L did not happen.
6. ✓ Observed — Hyperventilation kept until $EtCO_2$ has stayed below 40 mmHg for 2 min, then ventilation normalised (back to 12 × 500)
   The state "Ventilation back to 12 × 500, O_2 100 %" at 14:28, after $EtCO_2$ had stayed below 40 mmHg for 2 min 00 s (09:03 → 11:03).

Key moments and what the

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values at that second

n	Time	Who	What happened	HR	MAP	SpO_2	$EtCO_2$
1	00:34	Team	Midazolam 2 mg and fentanyl 100 µg given, breathing room air	70	94	96	37
2	01:06	Team	Preoxygenating: 100 % O_2 by face mask	65	90	96	38
3	04:05	Team	Induced: propofol 2 mg/kg, succinylcholine 1.5 mg/kg (masseter rigidity); airway not yet secured	63	89	99	46
	Alarms since the last row: RR low						
4	05:04	Team	Intubated: ventilator 12 × 500, PEEP 5, FiO_2 0.5; sevoflurane 2 % at 2 L/min	67	70	99	0
	Alarms since the last row: RR low, $EtCO_2$ low, Apnoea (no CO_2 breaths)						
5	06:05	Team	Surgery under way: sevoflurane 2 %, rocuronium	71	67	99	42
	Alarms since the last row: RR low, awRR low						
6	08:38	Patient	Early signs: $EtCO_2$ rising at the same ventilation, tachycardia	78	78	99	45
7	08:40	Team	Trigger off: sevoflurane stopped, O_2 100 % at 10 L/min, ventilation 20 × 600	80	78	99	45
8	09:00	Team	Dantrolene 2.5 mg/kg given	79	79	99	40
9	09:03	Patient	MH treatment under way	79	79	99	39
10	09:11	Team	Cooling: 2 L saline at room temperature running	81	80	99	38
11	09:13	Patient	MH treatment under way	80	80	99	38
12	10:13	Patient	Recovering: $EtCO_2$ below 45	81	85	99	36

1 more in the timeline

Measured facts

Lowest SpO_2	96 % at 00:10
Lowest ART M / NIBP M	62 mmHg at 05:25
Highest $EtCO_2$	54 mmHg at 20:57
Highest T1 / T2	37.2 °C at 25:12
Apnoea alarm to ventilated breaths	17 s (04:47 → 05:04)
Labs (the model): last (lowest–highest)	K^+ 4.6 mmol/L (4.2–4.8) · Lactate 1.6 mmol/L (1.0–1.6) · pH 7.28 (7.26–7.42) · Glucose 6.6 mmol/L (5.6–6.6) · Tcore 37.2 °C (36.6–37.2)

Talking points

Reactions
How did that feel? What stands out for you?

Description
In a sentence or two: what was the problem, and when did you know?

Analysis
Take one key moment: what did you see, what did you think, what did you do?

Summary
One thing to keep doing, one thing to do differently next time.

Notes

Kit round A4 rehearsal of the live-

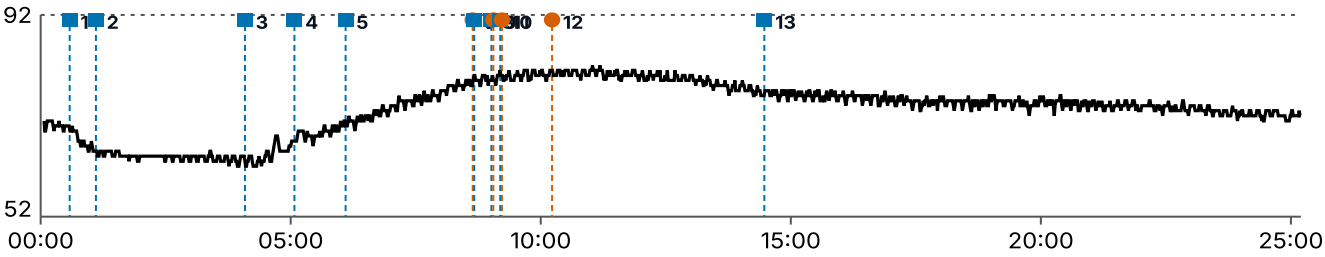
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Trends

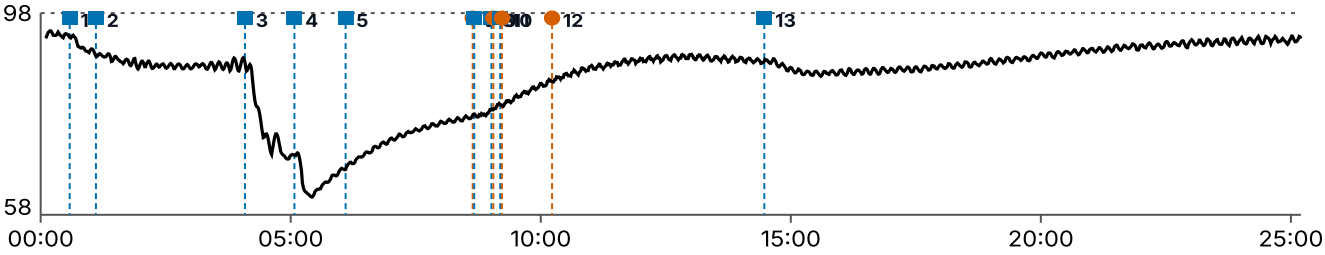
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■ team action · ● patient event · | scenario

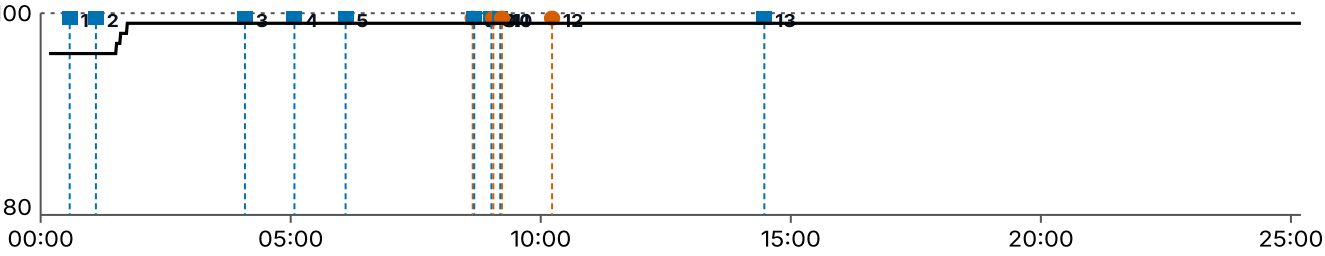
HR bpm



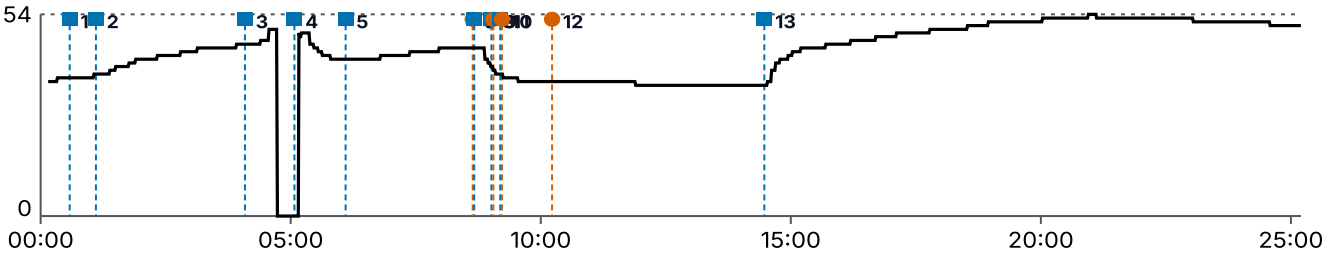
ART M / NIBP M mmHg



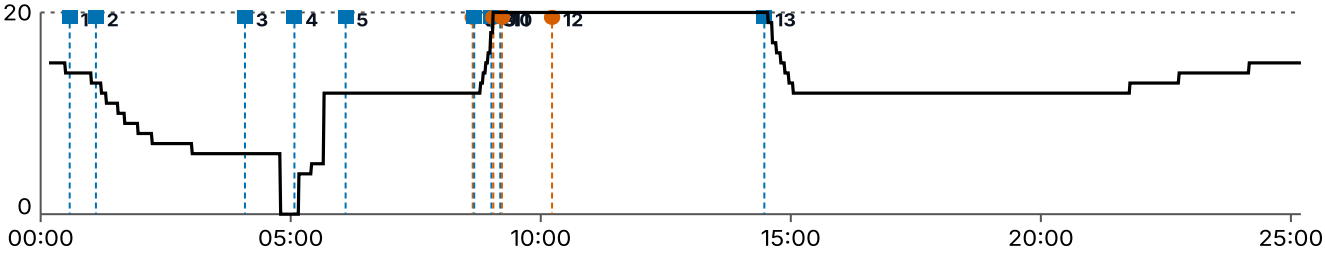
SpO_2 %

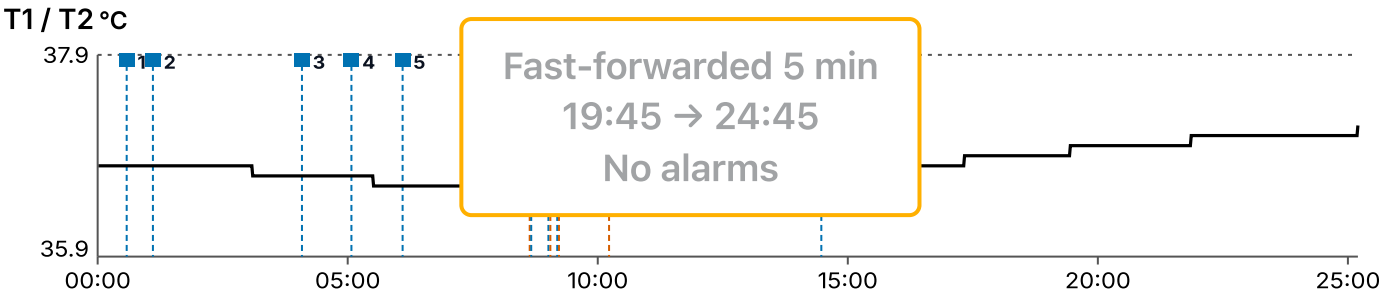


$EtCO_2$ mmHg



awRR / RR /min





Drugs and fluids

Time	Given
00:34	Midazolam 2 mg IV
00:34	Fentanyl 100 µg IV
04:05	Propofol 2 mg/kg IV
04:05	Succinylcholine 1.5 mg/kg IV
06:05	Rocuronium 0.6 mg/kg IV
09:00	Dantrolene 2.5 mg/kg IV
09:11	Crystalloid 2000 mL over 15 min

- Midazolam: 2 mg in 1 bolus
- Fentanyl: 100 µg in 1 bolus
- Propofol: 160 mg in 1 bolus
- Succinylcholine: 120 mg in 1 bolus
- Rocuronium: 48 mg in 1 bolus
- Dantrolene: 200 mg in 1 bolus

Timeline

Time	Kind	Event
00:00	Scenario	Scenario loaded: Malignant hyperthermia
00:00	Scenario	Scenario state: Awake on the table, monitoring on, breathing room air
00:01	Action	Speed ×8
00:04	Bookmark	Bookmark: Bookmark 1 at 00:04
00:34	Scenario	Scenario state: Midazolam 2 mg and fentanyl 100 µg given, breathing room air
00:34	Scenario	Midazolam 2 mg IV
00:34	Scenario	Fentanyl 100 µg IV
01:06	Scenario	Scenario state: Preoxygenating: 100 % O_2 by face mask
01:06	Scenario	Preoxygenation, FiO_2 100 %
03:07	Alarm	RR low
04:05	Scenario	Scenario state: Induced: propofol 2 mg/kg, succinylcholine 1.5 mg/kg (masseter rigidity); airway not yet secured
04:05	Scenario	Propofol 2 mg/kg IV
04:05	Scenario	Succinylcholine 1.5 mg/kg IV
04:05	Scenario	Malignant hyperthermia: severe

Time	Kind	Event
04:34	Alarm	RR low
04:45	Alarm	$EtCO_2$ low
04:47	Alarm	Apnoea (no CO_2
05:04	Scenario	Scenario state: Intubated: ventilator 12 × 500, PEEP 5, FiO_2 0.5; sevoflurane 2 % at 2 L/min
05:04	Scenario	Tracheal tube in place
05:04	Scenario	Preoxygenation, FiO_2 21 %
05:04	Scenario	Ventilator: RR 12 /min, VT 500 mL, PEEP 5 cmH_2O, FiO_2 50 %
05:04	Scenario	Sevoflurane 2 % at 2 L/min fresh gas
05:04	Patient	Ventilated breaths began
05:10	Alarm	RR low
05:10	Alarm	awRR low
06:05	Scenario	Scenario state: Surgery under way: sevoflurane 2 %, rocuronium
06:05	Scenario	Rocuronium 0.6 mg/kg IV
07:35	Action	Speed ×4
08:38	Scenario	Scenario state: Early signs: $EtCO_2$ rising at the same ventilation, tachycardia
08:40	Note	Objective met: Name malignant hyperthermia from the monitor: $EtCO_2$ above 45 mmHg at the same ventilation, tachycardia (instructor-assessed)
08:40	Scenario	Scenario state: Trigger off: sevoflurane stopped, O_2 100 % at 10 L/min, ventilation 20 × 600
08:40	Scenario	Sevoflurane vaporiser off
08:40	Scenario	Ventilator: RR 20 /min, VT 600 mL, PEEP 5 cmH_2O, FiO_2 100 %
09:00	Scenario	Scenario state: Dantrolene 2.5 mg/kg given
09:00	Scenario	Dantrolene 2.5 mg/kg IV
09:03	Scenario	Scenario state: MH treatment under way
09:11	Scenario	Scenario state: Cooling: 2 L saline at room temperature running
09:11	Scenario	Crystalloid 2000 mL over 15 min
09:13	Scenario	Scenario state: MH treatment under way
09:14	Action	Fast-forward 5 min
10:13	Scenario	Scenario state: Recovering: $EtCO_2$ below 45
14:28	Scenario	Scenario state: Ventilation back to 12 × 500, O_2 100 %
14:28	Scenario	Ventilator: RR 12 /min, VT 500 mL, PEEP 5 cmH_2O, FiO_2 100 %
14:31	Action	Fast-forward 5 min
18:33	Alarm	$EtCO_2$ high
19:45	Action	Fast-forward 5 min
24:59	Action	Simulation paused
24:59	Action	Simulation resumed

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Bookmarks

- 00:04 Bookmark: Bookmark 1 at 00:04

Appendix: one row per minu

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Time	HR bpm	ART M / NIBP M mmHg	SpO_2 %		/min			Lactate mmol/L	pH	Glucose mmol/L	Tcore °C
00:00	71	93	96	36	15	36.8	4.2	1.0	7.41	5.6	36.8
01:00	66	91	96	37	14	36.8	4.2	1.0	7.39	5.6	36.8
02:00	64	89	99	42	8	36.8	4.2	1.0	7.36	5.6	36.8
03:00	63	87	99	44	7	36.8	4.2	1.0	7.33	5.6	36.7
04:00	64	88	99	46	6	36.7	4.2	1.0	7.32	5.6	36.7
05:00	66	70	99	0	0	36.7	4.3	1.0	7.28	5.6	36.6
06:00	71	67	99	42	12	36.6	4.5	1.0	7.34	5.6	36.6
07:00	74	74	99	43	12	36.6	4.6	1.0	7.34	5.6	36.6
08:00	77	77	99	45	12	36.6	4.8	1.0	7.33	5.6	36.6
09:00	79	79	99	40	18	36.6	4.7	1.0	7.38	5.7	36.6
10:00	80	84	99	36	20	36.6	4.6	1.1	7.41	5.8	36.6
11:00	81	88	99	36	20	36.7	4.6	1.1	7.41	5.8	36.7
12:00	80	89	99	35	20	36.7	4.5	1.2	7.41	5.9	36.7
13:00	80	90	99	35	20	36.7	4.5	1.3	7.42	6.0	36.7
14:00	78	90	99	35	20	36.7	4.4	1.3	7.42	6.0	36.7
15:00	76	88	99	43	13	36.8	4.4	1.4	7.34	6.1	36.8
16:00	76	87	99	46	12	36.8	4.4	1.5	7.32	6.2	36.8
17:00	75	87	99	48	12	36.8	4.4	1.5	7.30	6.2	36.9
18:00	75	88	99	50	12	36.9	4.4	1.6	7.29	6.3	36.9
19:00	75	89	99	52	12	36.9	4.5	1.6	7.28	6.3	37.0
20:00	75	91	99	52	12	37.0	4.5	1.6	7.27	6.4	37.0
21:00	74	91	99	54	12	37.0	4.5	1.6	7.26	6.4	37.1
22:00	73	92	99	53	13	37.1	4.5	1.6	7.27	6.5	37.1
23:00	74	93	99	53	14	37.1	4.5	1.6	7.27	6.5	37.1
24:00	72	93	99	52	14	37.1	4.5	1.6	7.27	6.5	37.1
25:00	72	93	99	51	15	37.1	4.6	1.6	7.28	6.6	37.2